Siva Sankar Kaniganti

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SUMMARY

I am a **Machine Learning Engineer** with **2+ years of experience**, specializing in machine learning and deep learning algorithms. My expertise extends beyond technical implementation to clear, articulate explanations, demystifying complex concepts. I excel in end-to-end development within **startup environments**, tackling tasks from data preprocessing and modeling to **deploying solutions on AWS**. Known for developing custom techniques and deep learning architectures, I craft well-defined problem statements that optimize machine learning algorithms for state-of-the-art results. I stay updated with the latest advancements in deep learning, particularly in Generative AI, through continuous research. Additionally, I possess strong financial acumen, including evaluating financial instruments using various metrics

EXPERIENCE

Machine Learning Engineer

Sustainability Economics — **May 2023 - Present, Bangalore**

- Fine-tuned **multiple large language models (LLMs)** using **QLoRA** for efficient data extraction across diverse financial instruments and domain-specific energy and financial content. The models were designed to answer queries and resolve doubts, providing accurate, context-aware responses. Deployed the models using **vLLM**, enabling scalable, high-performance inference to support internal team workflows
- Developed an automated web scraping tool leveraging **ScraperAPI** for seamless data collection from various sources, including Google search results, blogs, PDFs, and Excel files. Integrated LangChain with the **Qwen2.5-Coder-7B-Instruct-GPTQ-Int8** model for context-aware processing and insights extraction. Designed custom tools, including **WebSearch**, **PDFReader**, and **ScrapeWebsite**, to build **agent-based scrapers** for handling complex, multi-format data sources. Implemented an in-memory **Retrieval-Augmented Generation (RAG)** mechanism to optimize the data acquisition process and supported research, competitive analysis, and content aggregation workflows for the internal team
- Developed a system to identify key details in Power Purchase Agreements (PPAs), such as project capacity, type (e.g., coal, solar, wind), term and duration, and pricing, through **extractive summarization techniques**. Achieved an **F1 score of 0.91**, accurately pinpointing relevant sections in diverse document
- Developed and deployed the **SEN Tool** for Sustainabiliteconomicsnews to automate the daily classification of over 4,000 news articles from various sources. Used **RoBERTa**'s **zero-shot classification** model to categorize articles into predefined categories, achieving **94% accuracy**. Integrated **Amazon SNS** for automated email notifications to journalists, reducing manual news distribution and improving the timeliness of publication.
- Developed a **Gradient boosting regression** model to predict carbon credit prices, ensuring interpretability through feature importance analysis and achieving **93% out-of-sample prediction accuracy**, enabling informed investment decisions in the carbon offset market
- Automated extraction of Scope 1, 2, and 3 emissions data from 5,000+ sustainability reports using **OpenAI models** with **vector database** employing **Retrieval-Augmented Generation (RAG)** architecture on **AWS EC2**. Achieved **95% precision** and reduced manual processing time by 90%, enabling timely and accurate carbon footprint analysis for climate risk assessment
- Developed a Python solution to bypass **Captcha** challenges and automate data extraction from a secure platform using **OCR** with **pytesseract** and **web scraping** techniques. Designed a pipeline to handle complex JSON data, preprocess it with Pandas, and store the results in structured formats like Excel.
- Developed a **ReAct** bot to handle user queries for financial calculations such as NPV (Net Present Value) and ROI (Return on Investment)
- Developed a machine learning-driven **optimization** model to balance renewable energy sources with demand, minimizing costs and grid stability.
- Built end-to-end ML products using **MLflow** and Docker on **AWS EC2,** incorporating **MLOps** practices
- Worked closely with and assisted the data engineering team in dividing financial data into different tables for **DynamoDB** loading from **S3**, utilizing **AWS Lambda**,**AWS BedRock** and **Glue**

Machine Learning Engineer

1stZoom — **November 2022 - April 2023, Bangalore**

- Worked on computer vision tasks, including camera tampering detection integrated into the CamBoss app using a Docker CI/CD pipeline.
- Developed abandoned object detection with **centroid-based tracking** and **Canny edge detection**
- Trained **YOLOv3** for object detection and integrated lightweight OCR models (**LPRNet**, **EasyOCR**, **Tesseract**) for efficient text extraction
- Built and deployed **Fast APIs** for image and video processing using **Docker** for scalability

AI Engineer - Internship

SkyGoal — **September 2021 - November 2021, Vijayawada**

- Developed a chatbot for exams and result-related information, designed to adapt to daily data changes with high speed and cost efficiency.
- Pre-processed data using NLP techniques and implemented **Word2Vec** for word vectorization.
- Overcame limited dataset size by employing **K-Nearest Neighbors (KNN)** to identify the most similar question and provide responses.

- Deployed the model using **Flask** on **Oracle Cloud**, ensuring smooth operation and accessibility.
- Integrated the chatbot with platforms such as **WhatsApp**, **Telegram**, and **Facebook**.

CERTIFICATIONS

Deep Neural Networks for Computer Vision and Natural Language Processing

IISc CCE • 2023

- Semester-long online courses offered by IISC CCE Data-driven modelling and Optimization, Advanced Topics in Deep Learning, Deep Neural Networks for Computer Vision and Natural Language Processing

Applied Machine Learning

Appliedaicourse • 2022

- Learned Basics of Mathematics, Dimensionality reduction techniques, NLP techniques, supervised and unsupervised Machine learning algorithms, Recommender Systems and Matrix Factorization, Deep learning architectures, Calibration of models, outlier removal techniques etc.

Practical Machine Learning with TensorFlow

IIT Madras • 2021

- This is an applied Machine Learning Course jointly offered by Google and IIT Madras. Learned Deep learning practically using TensorFlow

SKILLS

Languages: Python, SQl

Field of Interest: Machine Learning, Computer Vision, Ai, Deep Learning, NLP,GPT,BERT

Key Libraries & Frameworks: Scikit-learn, TensorFlow, Keras, Pytorch, PySpark, Flask, Selenium, NumPy, Pandas, NLTK,Regex, Streamlit, Huggingface, LangChain, FastAPI, FlaskAPi, Docker, CI/CD , MLFlow ,Ollama, webscraping, Playwright, OpenAI, Gemini, vLLM, TensorRT, llanggraph, llama-index, CrewAI, AWS lambda, AWS S3, AWS EC2, AWS SNS, Agentic RAG, Ragas, Vector databases, Chromadb, Pinecone, AWS Vectordb, AWS Bedrock

Machine Learning & Data Science: Proficient in ML, Computer Vision, NLP, Deep Learning, Data Analysis, Visualization, and WebScraping,RAG,LoRA, QLoRA,GPT,Bert,FlaskAttention. Hypothesis Testing, Distributions, PCA, TSNE, K-means, kmeans++, DBSCAN

Cloud: Amazon Web Services

PROJECTS

Developed and trained a GPT model from scratch using Python and PyTorch frameworks, with a focus on next character generation

Implemented Machine learning techniques from scratch

- Logistic Regression using SGD with L2 Regularization, Backpropagation using several optimizers like Adam, RMSprop, Vanilla SGD, SGD with momentum for a small neural network,Calibration with regularization (Platt scaling), Bootstrap using Decision trees, Recommendation systems and Truncated SVD SGD algorithm, Performance metrics , RandomSearchCV with K fold , TFIDF vectorizer, etc.. were implemented just using numpy

EDUCATION

B.TECH Computer Science and Engineering

Sri Chandrasekharendra Saraswathi Viswa Mahavidyalaya • Kanchipuram • 2022